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Integrated fire management in a tropical biosphere reserve: Achievements and challenges

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ABSTRACT

Wildfires are among the most critical threats to tropical rainforests. In the Selva El Ocote Biosphere Reserve in the Mexican state of Chiapas, vast rainforest areas suffered destruction by fires in 1998 and 2003. However, there have been no more widespread fires, and previously burnt areas have been recovering. The present study analyses the integrated fire management processes implemented in the reserve. It includes 19 expert interviews and a survey applied to 274 inhabitants of local communities. It argues that the successful prevention of further adverse wildfire events is related to a fire management approach, which emphasizes cultural and physical fire prevention and incorporates strong participation by local communities. The communities have established internal agreements to mitigate wildfire risk. Thus, the reserve might serve as an example of how integrated and cultural fire management can be applied in practice. However, these achievements face challenges, mainly due to budget cuts, changing governmental priorities, and the crisis of the rule of law in Mexico. An event that particularly impacted fire management practices in the area in recent years was the implementation of the federal government's *Sembrando Vida* ("Planting Life") program. First, it led to deforestation and the provocation of wildfires but afterwards it seems to have induced the abandonment of agricultural burning by a considerable percentage of local farmers.

1. Introduction

Wildfires are an essential element in the natural dynamics of many ecosystems. Humans have used fire for millennia, particularly in swidden agriculture. However, nowadays, fire regimes have been altered in most parts of the world, with excessive fire suppression in some fire-prone areas and too much fire in fire-sensitive vegetation, especially in the tropics [1]; in some areas with generally fire-adapted vegetation, fires have become so frequent or intense that it also affects the ecosystem negatively [2,3]. This jeopardizes the conservation of ecosystems and biodiversity [4] and contributes to climate change by releasing greenhouse gases and degrading forests acting as carbon sinks [5,6]. Through their environmental impacts, destructive wildfires diminish the provision of essential ecosystem services such as water, wood, and non-wood forest products [7]. Moreover, fire affects human communities directly, destroying crops, dwellings, and infrastructure, contributing to air pollution, and endangering human lives [8,9].

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The response to these impacts has generally entailed a one-sided focus on fire suppression, which has brought about several problems: (1) negative ecological consequences and even an increased risk of uncontrollable megafires in areas with fire-prone vegetation [1,10,11]; (2) the abandonment of traditional fire risk management practices and cultural knowledge [11,12]; and (3) adverse effects for peasants whose traditional farming involves the use of fire [8,13].

1.1. Integrated fire management

This paper follows an *integrated fire management* (IFM) perspective, a concept developed in the past two decades that is opposed to conventional response-only focused approaches to fire and avoids total fire exclusion, promoting the strategic use of fires in prescribed burning [14,15]. In some cases, it also allows traditional fire use. In short, IFM aims to decrease fire damage while maintaining or even increasing the beneficial effects of fire [16].

Nevertheless, even under the label of IFM, sometimes restrictions for local farmers prevail, and solutions are rather top-down, with scant regard for the opinion of local inhabitants [11,13,17]. Also, some conceptualizations of IFM still prioritize response-focused approaches, resulting in, as Tedim et al. state, just a way to improve the conventional "war on fire" model instead of a real paradigm shift [18].

Bacciu et al. [19] are less critical of the IFM concept, characterizing it as combining prevention and suppression strategies, prioritizing fuel management, and highlighting key stakeholders, particulary local communities. However, they see it as one out of three main approaches to fire management that entail more than mere response; the others are (1) *social-ecological systems*, prioritizing human-environment interactions at multiple scales and sustainable coexistence with wildfire; and (2) *fire resilience*, involving concepts of risk, adaptation and mitigation in the context of wildfires, also containing the concepts of *firescapes* and *fire smart territories*. Nevertheless, the authors acknowledge interrelationships between these approaches; for instance, studies they assign to the IFM category that consider a resilience perspective [20]. Finally, they propose an integrated (defining this term as "multi-level, multi-actor, cross-sectoral, and multi-purpose") and adaptive approach instead of "silos" that only consider specific aspects.

The present study follows an IFM approach in a broad sense, "bringing together diverse concepts (…) synergistically" [16] and not excluding any of the approaches that are opposed to the conventional emergency response paradigm, coinciding, for instance, with Wunder et al. who see their resilience approach as being part of IFM [15]. However, the study sets specific priorities by emphasizing the role of stakeholders and their management practices (for prevention, preparedness, response, and recovery), following Moore [21], and the fire-related practices of local communities, according to Myers [16]. It subdivides preventive activities in cultural prevention (oriented towards avoiding accidently caused fires by the population), physical prevention (technical activities, such as opening fire breaks), and legal prevention (laws and decrees regarding fire use and arson, and their implementation), and divides response in detection of fires and fire suppression [22].

Under this conception, also the insights of several more recent approaches can be integrated, such as *cultural fire management* (prioritizing fire use and management at the community level related to the locals' knowledge and experiences and community self-organization [23]) or *intercultural fire management* (as a general management scheme where local traditional practices are included and local inhabitants are part of decision-making processes [11,24–26]), or *shared fire governance*, focused on empowering community engagement, addressing human fire causes, reducing vulnerability, losses, and civilian and firefighter safety risks, as well as enhancing ecological benefits of fire, effective response and recovery [18].

Stakeholder-focused approaches to research in this field have brought about important insights. For instance, researchers claim that it is vital to change the general narrative of small-scale farmers as the main initiators of large, destructive wildfires, ignoring fires caused by large-scale industrialized agriculture, logging, and mining [8,27]. Furthermore, studies have highlighted the importance of traditional management practices to prevent adverse fire outcomes [24] and the crucial role of local persons as professional or voluntary firefighters and of organization at the community level [22]. They have also shown that the erosion of local knowledge due to top-down policies, migration, and tourism can increment fire risk [28]. The importance of this traditional knowledge is especially highlighted within the *fire smart territory* framework, which stresses that management solutions need to be based on an understanding of the specific local human and natural systems of each area [29].

In general, Smith et al. call for avoiding "generic 'oppositional' perspectives that pit 'locals' against state or non-governmental organizations" but for research that acknowledges and revises the co-production of knowledge and negotiations between stakeholders, wherein interpersonal relationships are crucial [11]. Moreover, Weber et al. stress that discrepancies in perceptions, attitudes and behaviours regarding fire exist on different levels, not only between authorities and landholders but also within social groups, for example, between genders [30].

However, despite these advances, there are gaps in previous research and aspects that have received less attention. Globally, most studies so far have focused on industrialized countries and ecosystems with a Mediterranean or temperate climate and a predominance of fire-prone ecosystems. Furthermore, studies often regard a relatively short temporal scale and tend to be centred on cases where fire management does not work well [2,8,10,13,17,27] and on recommending better new strategies [18–20,29,31]. On the contrary, there are only a few exceptions where the implementation of good practices over a longer term has been documented [24,25,32] that could be adapted in other areas, showing a way for a sustainable human-fire relationship. In the words of Wunder et al. there is still no "proven track record" of alternative fire management strategies [15].

1.2. Outline of the study

The present paper aims to address this research gap and deepen the understanding of IFM concerning a case study in a protected area in the tropics, the Selva El Ocote Biosphere Reserve in Chiapas, Mexico. The selection of this considers the following aspects: First, the area is of high importance for conservation at the national and global scale, given its biodiversity and ecosystem services;

second, its vegetation is mainly made up of fire-sensitive tropical rainforests, which suffered from large-scale destruction due to huge mega-fires in 1998 and 2003 [33,34]; the geographical background and the impacts of these events are described in detail in chapter 2. Third, to understand the role of fire in the reserve, it is necessary to consider that 60.1 % of the total 101,288 ha that are protected are a buffer zone that includes human communities (some being mestizos and others being indigenous Tzotzil), who traditionally use fire in their agricultural activities [33]. Fourth, wildfires remain a significant threat to the area's tropical rainforests, but adverse fire impacts have decreased considerably. This last consideration also indicates that the way fire management has been applied in the area might somehow have influenced this positive tendency and might follow general IFM principles, either on purpose, or developed organically by local actors based on their own experiences.

Based on this initial panorama, this paper records how fire prevention and suppression have been applied within the reserve in relation to local burning practices and fire risk management in the communities within the buffer zone. The concept of IFM, as described previously, serves as a framework to guide this analysis. The paper aims to identify the practices and successful outcomes of fire management, particularly by considering principles of IFM, in the Selva El Ocote Biosphere Reserve and pinpoint the problems encountered so far. Responses to interviews and questionnaires reveal the experiences, activities, and perceptions of the actors (representatives of governmental and non-governmental organizations who work in the reserve, local authorities, local firefighters, and the general population of the communities in the buffer zone). Thereby, the study seeks to identify processes and structures required to implement IFM.

2. Context of the case study

The Selva El Ocote Biosphere Reserve occupies an area of 101,288 ha in the northwestern part of Chiapas, Mexico. Conservation efforts had begun in the 1950s, and since 1982, there had been a protected area of 48,140 ha [33]. This culminated in 2000 with the official declaration of the biosphere reserve, administered by the National Commission of Protected Areas (CONANP). This federal administration coincides with other administrative authorities: the state government of Chiapas and the governments of four municipalities that share parts of the reserve (Cintalapa, Jiquipilas, Ocozocoautla de Espinosa, Tecpatan) (Fig. 1). Land tenure consists of private cattle ranches and of community land dedicated to agricultural activities. The latter is organized in the *ejido* system, ruled by a community assembly in which all those who share land ownership rights participate.

The main reason for establishing this reserve was to conserve its rich biodiversity, holding one of Mexico's last remnants of tropical rainforest. Many of the species found here are classified as endangered, and several are endemic. Moreover, the forests of Selva El Ocote, growing over an area characterized by karst, are crucial for capturing water [33]. The last detailed study of the area's vegetation dates from 2009 [40]. That year, 75.0% of the reserve was tropical rainforest, an ecosystem highly sensitive to fires, with natural occurrence rates of centuries or millennia [41]. Other kinds of natural vegetation (savannah, oak forest, low tropical, deciduous forest, and medium semi-evergreen forest) made up 9.1%, whereas 9.3% were fields used for agriculture and 5.9% were pastures) [40]. Fig. 1 gives an overview of vegetation and land uses in the reserve based on more recent but less exact data [36].

In biosphere reserves, a category initially developed by UNESCO, the aim is not only to conserve nature but to do so in harmony with the sustainable development of the local communities [42]. Hence, whereas the most conserved parts of the reserve are core zones with restricted access, 60.1% of the area was established as a buffer zone, in which land uses such as settlements and agriculture are permitted. Within the buffer zone are several subzones dedicated to distinct uses, including a traditional use zone [33].

According to the most recent census from 2020, 9397 people were living in the reserve, mainly in small communities (primarily organized as ejidos) with a maximum population of 861; 63.0% of the inhabitants were members of indigenous households [43], mainly Tzotzil. Most of the local workforce engages in primary activities, especially subsistence maize production, depending directly on the reserve's ecosystem services. Different kinds of uses of forest resources, such as wild fruits and firewood, are widespread. Many farmers also cultivate coffee, which grows under the shade of forest trees.

The use of controlled burns in agriculture is a traditional practice in the communities [33]. In general, the reserve's management plan considers traditional practices (a part of the buffer zone is defined as a subzone for traditional use) of both indigenous and non-indigenous communities but imposes restrictions as well, for instance, with regard to the use of non-wood forest resources, which may conflict with the local communities' needs [44].

The Biosphere Reserve faces many threats, such as deforestation related to the expansion of the agricultural frontier, illegal logging and hunting, and uncontrolled tourism [33]. However, the most severe threat is wildfire, which is particularly difficult to combat since the karst topography causes many fires to be subterranean [34]. The large areas of secondary vegetation pictured in Fig. 1 are related to the widespread fires of 1998 and 2003. These large-scale fires appear to have been an entirely new phenomenon in the area because there are no records of previous events of that magnitude; their causes are unknown, but the local inhabitants interviewed for this study generally assume that they were caused accidently.

The estimate of the area burned differs according to sources. For 1998, the reserve's management plan estimated 19,000 ha [33], whereas analysis of satellite data suggested 31,997 ha [38]; in addition, another study based on satellite data [39] found a burned area of about 10,000 ha in the north of the Cintalapa municipality, mainly within the reserve, which for the most part had not been detected by Manzo-Delgado et al. [38]. For 2003, official records accounted for 9275 ha, whereas only 1301 ha could be detected with satellite images, owing to the density of cloud cover [38].

There is no more specific information on the ecosystems affected. However, all cited sources agree that in the main part it was primary vegetation of high tropical evergreen forests that burned. Other types of vegetation have been less affected. There are no specific studies on this issue regarding Selva El Ocote, but in other parts of Chiapas, oak forests are shown to be fire-dependent [45], and low tropical deciduous forests can adapt to fires [46]. Medium semi-evergreen forests in Mexico are usually not fire-adapted, while savan-

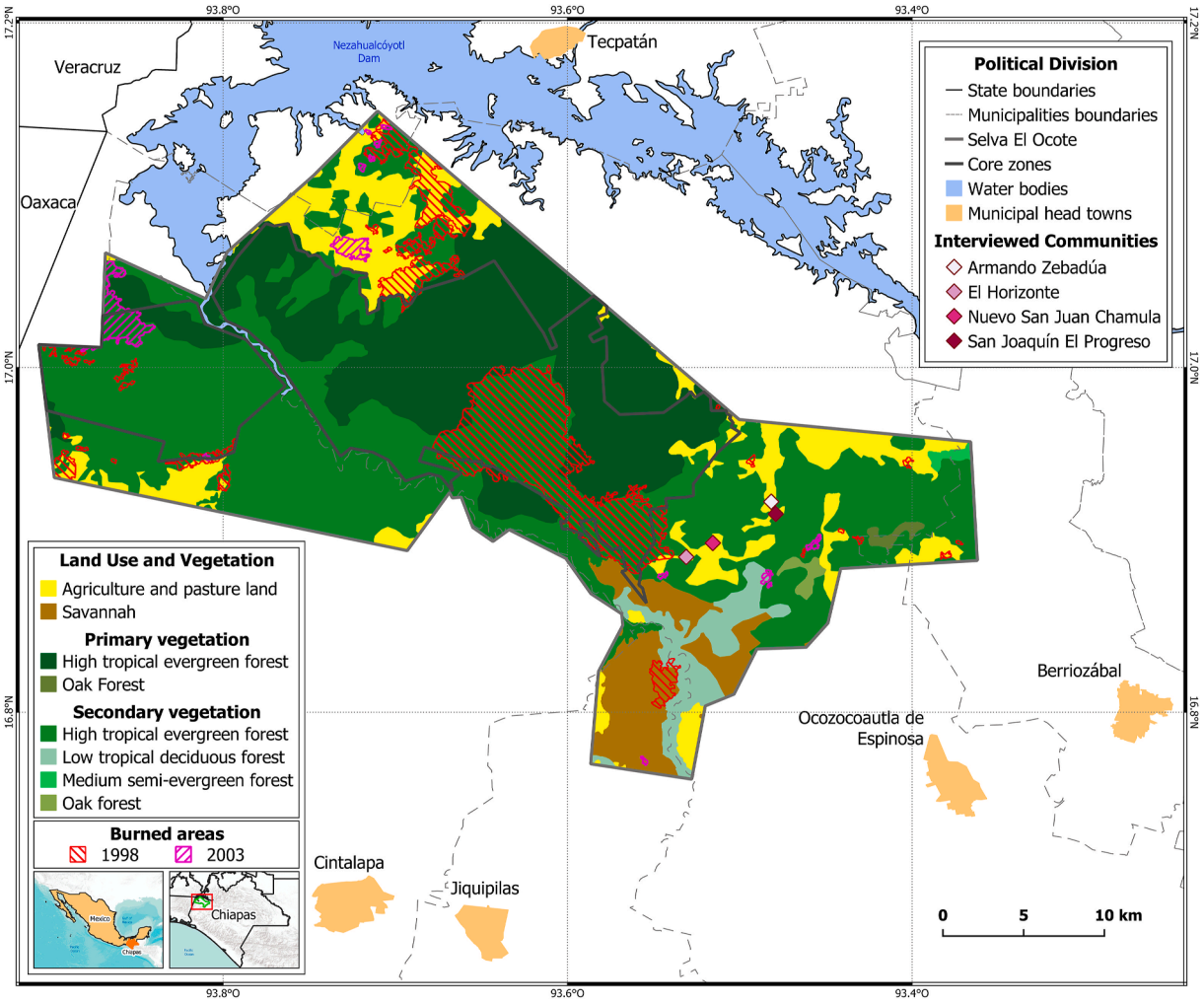

Fig. 1. Land use and vegetation in 2016 in the Selva El Ocote Biosphere Reserve and extent of the 1998 and 2003 wildfires.
Source: Based on Comisión Nacional de Áreas Naturales Protegidas [35] and Instituto Nacional de Estadística y Geografía [36,37] and wildfire data classified by Manzo-Delgado et al. [38].
Note: The wildfire data shown here are incomplete, owing to problems with detection by remote sensing methods. In 2003, the number of burnt areas found by Manzo-Delgado et al. [38] was much lower than that of official data. In 1998, another study based on satellite images [39] and focused only on the part belonging to the municipality Cintalapa, in the Southwestern part, found that almost all the natural vegetation there was burned, whereas the results from Manzo-Delgado et al. suggested only minimal impacts in that area.

nahs generally depend on fire [41]. In the case of Selva El Ocote, Maldonado et al. indicate that the savannahs are of anthropogenic origin, growing on areas that were previously tropical forests [34].

The 1998 fires caused 39 thousand tons of carbon emissions per hectare and soil erosion of 34 tons per hectare. Subsequent studies documented the effects on the habitat of endangered species such as jaguar *Panthera onca*, mantled howler *Alouatta palliata*, and Central American spider monkey *Ateles geoffroyi*, and the loss of tree species diversity and tree cover in burnt areas, especially in those affected repeatedly; they also showed that rainforest that had burnt once is more prone to future fires [34,41].

Since then, the incidence of large wildfires has decreased, although there are still annual wildfires in the area. Between 2004 and 2012, for instance, the yearly average, according to official data, was 382 ha [38]; the leading cause of the wildfires in the area according to the management plan is agricultural burning, although there are also wildfires caused by lightning, by hunters, and by cigarette butts [33].

The reserve's management plan contains a specific subcomponent for wildfire prevention and control [33]. In theory, the management plans of protected areas in Mexico result from a participatory process. However, Zamora-Lomelí [44] interpreted the initial implementation of the Selva El Ocote Biosphere Reserve rather as an imposition with little voice given to the local communities. Regarding fire use, the plan does not prohibit it, but is aimed at reducing it as much as possible. Since the plan was published in 2001, its relevance for the current implementation of management activities is limited; these instead rely on ad hoc planning and the experience acquired by the people employed in the reserve's administration, as was mentioned in the interviews with local actors mentioned in the following section.

3. Materials and methods

The main part of this research used three field visits to the study area: in November 2021, March 2022 (at the beginning of the wildfire season) and July 2022 (after the season). There were 19 semi-structured interviews with individuals and groups involved in fire management, from diverse backgrounds (Table 1). These key informants included representatives from all the relevant governmental and non-governmental groups identified, as well as a small sample of members of the communities within the reserve (*ejido* commissioners and local firefighters), which served especially as a preparatory step for a subsequent questionnaire in these communities.

The interview questions (see Supplementary material 1) were based on the literature on IFM and previous experiences of the research project participants in other study areas. Besides collecting general data, they were related to the research objective of documenting and analysing how fire management has been implemented in the reserve, its successes, and its problems. Thus, the interview guide included general questions about the interviewees' views on the fire situation in the area, followed by questions about their opinions regarding agricultural fire use. Furthermore, a large part of the interviews focused on fire management activities applied by the interviewees' organizations and by other actors, with specific questions on cultural, physical (including prescribed burning), and legal prevention, fire detection and suppression and post-fire restoration. The interviews closed by turning again to more general questions, such as the interviewees' outlooks on future developments and what they considered their main achievements, challenges, and learnings. Owing to the low overall number of organizations involved in the reserve, we reached theoretical saturation with these interviews regarding the work and perceptions of external actors. Regarding the communities, a questionnaire was added to explore perceptions at the local level further.

The semi-structured interviews were accompanied by participatory observation in capacity-building workshops with local firefighters and talks given by authorities to local communities as part of cultural prevention. The selection of these activities for participatory observation depended on two main aspects: the invitation by local actors and their temporal coincidence with field visits. Participation yielded first-hand experience of the situation in the area, built confidence among local actors, and allowed observation of stakeholders' interactions.

Furthermore, a household questionnaire used mainly closed questions (see Supplementary material 2) based on the information derived from the previous interviews and the participatory observation. The questions were again related to the overall research objective and touched on topics similar to those in the interview guide but with a much stronger focus on agricultural fire use and observed fire impacts. On the contrary, questions on fire prevention and suppression activities were rather general, supposing that most people did not participate actively.

The results of this data acquisition methodology provided data directly related to the study's research objective, documenting, for example, people's perspectives on the successes and problems of fire management in the area. This applied to all three data collection instruments (interviews, observations, and data from the questionnaires), documenting, for instance, the way traditional fire use has been managed and how community members have been involved in fire management activities and have received capacity-building in this regard. Furthermore, the ample format of the questions and the extensive nature of the questionnaires allowed it to gain contextual data that permitted a deeper understanding of the study object.

Representatives of governmental and non-governmental organizations working in the area reviewed the questionnaire. Subsequently, it was pretested with seven local firefighters. Any ambiguities were resolved by discussion with each pilot survey participant.

The resultant questionnaire was administered to 28 inhabitants of one community, and the results revealed no further problems. Consequently, the data from these 28 respondents were integrated into the overall survey dataset. In total, the questionnaire was distributed to 274 persons aged 18 years or older residing in four communities within the biosphere reserve's buffer zone (the area around the core zones), belonging to the municipality of Ocozocoautla: Armando Zebadúa, Nuevo San Juan Chamula, El Horizonte, and San Joaquín El Progreso. The communities have a mixed population of indigenous and non-indigenous inhabitants; most families depend on agriculture (Table 2).

The study was restricted to communities within the reserve because these are the ones in direct proximity to the reserve's forest areas. The survey excluded individuals residing within the core zone because there were potential security risks associated with survey administration (see Results section). It also excluded individuals living on or owning private ranches because of the challenge of contacting this scattered population and because unauthorized access to private properties would have raised security considerations.

Table 1
Overview of the interviews (including final focus group discussions).

Organization	Interviews	Interviewees
National Commission of Protected Natural Areas (CONANP)	5	9
National Forestry Commission (CONAFOR)	4	5
State Secretary of Environment and Natural History (SEMANH)	3	3
State Secretary of Civil Protection	2	2
Ambio Cooperative	1	1
Autonomous Agrarian University Antonio Narro (UAAAN); formerly member of Pronatura	1	1
Community-based firefighters	4	19
Ejido commissioner (main authority within the local communities)	3	3
Total	**19***	**43**

Note: * Two of the interviews were shared group interviews with members of CONANP, CONAFOR and SEMANH.

Table 2
Characteristics of the participants of the survey.

Aspect	Metrics
Responders per community	Armando Zebadúa: 86; El Horizonte: 69; Nuevo San Juan Chamula: 91; San Joaquín El Progreso: 28; TOTAL: 274
Ratio women:men	1:1
Origin of the responders	70.1% from the four communities, 14.9% from other places within the municipality Ocozocoautla, 15% from other places in Chiapas
Age	18–29 years: 39.8%, 30–64 years: 56.2 %, >64 years: 4%
Speakers of Tsotsil	78.8 % of the responders
Position within the community	Landowner: 29.6%; spouse of landowner: 14.2%; others: 56.2%
Employment	Housewife: 48.9%; farmer (*campesino*): 47.8 %; others (mainly students and teachers): 3.3 %

The selection of the participants was random, with one exception: in line with previous experiences of the persons involved in the study, an effort was made to obtain an equal number of male and female respondents. A fully randomized survey might have led to a disproportionally high response rate of women since men tend to be away from home for a large part of each day, engaged in agricultural work. The sample frame, based on official census data, underwent review with interviewees, including firefighters residing in the communities. It was assumed that, aside from the gender-related participation issues, no other factors might introduce sampling errors, because communities tend to exhibit a high degree of homogeneity regarding economic activities and social status. Consequently, coverage errors were minimized. Refusals to participate in the survey were minimal, occurring in only 6.8% of 294 households contacted. This rate includes cases where residents did not answer the door or indicated a lack of time; these cases were excluded from the general survey calculations.

The four communities were selected to represent the diversity of localities in the area in terms of size, presence of indigenous and non-indigenous population, incidence of wildfires, and use of controlled burns. Survey administration involved traversing routes within the communities and stopping at predetermined intervals of households, the frequency of which depended on the community's size.

Given the qualitative nature of this study, the survey's primary objective was to gain a comprehensive understanding of the range of experiences and perceptions among residents concerning the issue of fire within the reserve. Hence, descriptive statistics were employed to summarize and characterize the data instead of formal statistical tests. The results were presented in percentages related to the total sample or the number of respondents from one of the four communities. In the third and final visit, the preliminary findings of the survey were presented to two focus groups for discussion: the employees of the biosphere reserve (eight participants) and the local firefighters of Armando Zebadúa (seven participants).

Data analysis required transcription of the information derived from the interviews, focus group discussions and observations, and the application of textual analysis with a coding system to identify and order the most relevant issues. This coding was mainly deductive, with an initial code list based on the literature regarding IFM; additional codes were derived from other topics that had arisen in at least three interviews and seemed important for fire management. The data from the questionnaires was integrated into spreadsheets and visualized in diagrams to facilitate its interpretation. Subsequently, the final analysis triangulated the data from these different sources. It consisted of combining the codes applied to the interview transcriptions and the notes from participatory observation regarding the same issues and relating them to the questionnaire responses. This allowed it to obtain a more profound and complete understanding and to summarize the results in a figure.

4. Results

4.1. Wildfire causes and impacts

According to the data provided by the reserve's administrative office, in recent years (2017–2021), on average, 86.4 ha were burnt each year by uncontrolled fires within the reserve's boundaries. This reduction compared to previous years (see Chapter 2) is remarkable, given that at the same time, at the state level in Chiapas, there was an upsurge in burnt areas in federally protected areas, related to meteorological factors (according to several interviewees), with more than 23 thousand ha burnt in 2019, the highest number in ten years, and again almost 19 thousand ha in 2021. Among the areas with high numbers of burnt surfaces were also some relatively close to Selva El Ocote, like La Sepultura, and La Frailescana (data provided by the regional office of the CONANP). In only 17.8% of the burnt areas in Selva El Ocote were fires so intense they affected adult trees. Most fires were registered in tropical evergreen or semi-evergreen forests (68% of the burnt areas), whereas the remainder happened in areas with low tropical deciduous forests. The official data contain no further information on the fires' impacts on the environment or on society.

In the interviews, some environmental impacts were reported, mainly related to the large fires of 1998, when many animals were seen fleeing from the burnt areas, and the disappearance of some species, such as the harpy eagle *Harpia harpyja*, was related to these events. In recent years, firefighters have again witnessed mammals and birds fleeing from fires and reptiles burnt in the flames, but these events have been localized without further effects on the ecosystems as a whole. In response to an open question in the survey, 10.9% said that they knew of animals having died directly due to wildfires, and 4.7% recorded the migration of fauna to unburnt areas. Only 14.6% said that recent wildfires had destroyed forest areas. A further problem mentioned by the fire management leader of the reserve's administration was the tendency of the secondary vegetation that has grown on previously burnt rainforest areas to accumulate high amounts of combustibles, which poses an increased risk of a future mega-fire.

Regarding direct effects on humans, community-based firefighters were aware of the adverse health effects of fire smoke and reported having felt how it "dries the throat". In the survey, 24.8% of the respondents also mentioned the negative effects of wildfire smoke. 3.8% of the farmers noted that their crops had been destroyed by uncontrolled fire on their land, mainly due to the large fires in 1998. In terms of ecosystem services, one interviewee noted perceived water shortages, potentially related to forest degradation by wildfires. This aspect was also mentioned by 6.9% of the survey respondents, while 10.9% stated that wildfires "contaminated" water.

In the literature cited in the introduction, wildfires in the reserve were primarily related to fire use by farmers. In the present survey, only 69.7% of the surveyed population responded to this question. Of these, 64.6% named burning of cropland as the main cause, and 29.3% the use of fire on the private cattle ranches. Cigarettes were blamed by 4.7%.

According to the data from the reserve's administration, in recent years (2017–2021), 42.4% of wildfires were caused by agricultural burns, 21.2% by hunting (to rout out green iguana *Iguana iguana*, mainly undertaken by people coming from outside the reserve), 15.2% by burning on cattle ranches, and 6.1% by lightning strikes; in 15.2% of cases, the cause was unknown.

4.2. Integrated fire management

The following paragraphs deal with the aspects of IFM and their implementation in the reserve, documenting the strengths and achievements as well as the weaknesses and drawbacks of these approaches. Furthermore, they identify the main actors involved; Fig. 2 summarizes this information.

4.2.1. Cultural fire management

The survey confirmed the widespread use of agricultural burning in the reserve but showed noteworthy changes in recent years. Most of the farmers cultivate maize (93.9%). Other important crops are coffee (68.2%) and fruit trees (mainly citrus fruits) (20.4%). 4.6% also engage in silviculture, and 5.3% in cattle ranching. The total share of cattle ranching in the reserve is much higher (see Fig. 1), but according to the interviewees is mainly carried out on private ranches owned by people living outside the reserve. 54.8% of farmers cultivating maize used controlled burns, 54.5% of farmers in general said they used controlled burning, and another 40.9% of all farmers had done so in the past. The remaining 4.5% claimed never to have applied agricultural burning. Of those who had abandoned fire use, 83.7% had done so in the last five years.

In general, farmers used fire to clear stubble before planting and, in only two cases, to collect honey from wild bees. None of the responders used fire to burn pasture before replanting, a practice widespread in private ranches, according to several interviewees. As benefits of burning, fire users mentioned clearing the fields clean for seeding (58.3%), elimination of plagues (38.9%), improved sprouting (29.2%), higher yields (11.1 %), lower workloads (4.2%) and avoidance of agrochemicals (4.2%). The number of original fire users was almost the same among indigenous (95.4%) and non-indigenous people (95.9%). However, many more indigenous farmers (60.2%) still use fire today than do non-indigenous people (29.2%).

One of the contextual questions of the survey included subsidy programs that farmers participate in. The by far most widespread was the governmental program Sembrando Vida ("Planting Life") introduced in 2019, whose aim is to incentivise the implementation of agroforestry systems to reactivate the rural economy and counteract environmental degradation [47]. 52.8 % of the participating farmers were affiliated with this program. Also, in the interviews, when asked about fire use in the reserve, several respondents mentioned this program.

The interviews revealed a contradictory impact of this program regarding fire use and wildfire incidence in the reserve. At the beginning, during the program's initial implementation, there were increased intentional fires to burn vegetation and clear land of trees to comply with the conditions for entering the program (only plots free from any vegetation were accepted), sometimes causing uncontrolled wildfires. While some farmers did this secretly, in other cases, contrary to the program's aims, this practice was allegedly encouraged by the governmental agents facilitating its implementation. One community in the reserve decided collectively not to participate in *Sembrando Vida* to protect their forests and reforested areas. Later, the way the program was implemented turned to the opposite, prohibiting fire use at all, even in a controlled form. Thus, 63.4% of the survey respondents affiliated with the program claimed they had stopped using fire. 24% said they still used fire, although this might relate to parcels not registered within the program.

Of all those surveyed in this study who were not using fire, 51.9% stated they had stopped using it because of *Sembrando Vida* and 11.1% had done so because of unspecified prohibitions (open question without predetermined answers). Some respondents mentioned further reasons for not using fire, mainly related to the conservation of soil and its fertility (28.2%) and the conservation of fruit trees (5.6%). 5.8% said that after having followed the program's directive to stop using fire, they had seen that controlled burns were unnecessary. The adverse effects of fire on soil fertility were particularly mentioned by those who had stopped using fire a long time ago.

All farmers who still use fire, except one, mentioned measures they apply to avoid causing uncontrolled fires (open question without predetermined answers). The most frequently mentioned technique was the creation of fire breaks (91.7% of the respondents), while only 20.8% said they burn when it is not too dry (9.7% related this to burning in the morning and 4.2% to burning in the evening, while 1.4% wait for the first rains after the dry season), and 4.2% when there is not too much wind. Furthermore, 6.9% indicated that sufficient people accompanied them to put out any fire that might escape (between four and six additional assistants), and 4.2% said that they guarded the fire until it was completely extinguished.

Farmers learned these measures mostly from parents and other family members (76.4%) or their own experience (11.1%). In the interviews, respondents detailed that there are agreements by the community assemblies on how to use fire. In the survey, 75.2% said they knew this, 22.6% said they did not, and 2.2% stated that no such agreements existed. Most of those answering with "no" or "I do not know" were people not directly involved in agriculture (housewives, teachers, and students).

Fig. 2. Integrated fire management in the Selva El Ocote biosphere reserve.

When questioned on the community agreements, 53.2% mentioned the obligation to establish fire breaks before burning, 8.3% referred to other precautions such as avoidance of dry or windy conditions, 4.4% said that it was necessary to monitor the fire until it was extinguished and 2.0% said that the presence of assistants was obligatory. 39.5% stated that people who burn need to inform the local authorities or firefighters, and 25.4% said the agreements also implied sanctions. In an interview with local firefighters, the sanctions were said to include compensation for damage and payment for the labour of those who put out the fire. Several of the employees of governmental agencies, both in individual interviews and focus group discussions, underlined the importance of these community agreements and their effectiveness in contrast to external intervention. In the community assemblies, the local population also use to decide on their involvement in further fire management activities.

4.2.2. Cultural prevention

In addition to learning from family members and their own experience, 12.5% of the surveyed farmers said they had benefitted from external advice. However, in another survey question, most fire users (65.3%) said they had attended talks on safe burning techniques: in 59.6% of the cases, these were given by the CONANP, in 19.1% by a non-governmental organization called Ambio, in 14.9% by community authorities and in 5.6% by local firefighters. Although, according to the survey, attendance at talks did not aug-

ment the percentage of those applying safe burning techniques such as fire breaks, the local firefighters and governmental agents stressed the importance of these talks, especially in maintaining a dialogue with the local communities.

As with the rest of the formal fire management activities in the area, the talks started after the large 1998 fires, initially supported by the now absent organizations The Nature Conservancy (United States) and Pronatura. The talks mainly take place at community assemblies. Although the people who do not own land do not have the right to vote in these meetings, they attend to be informed. Thus, 96.0% of all survey respondents identified these meetings as the acceptable format to diffuse information on fire management, while 2.9% mentioned information on the radio and 1.1% on TV channels; only 0.4% mentioned information shared on the internet, since connectivity in the area is low. Once a year, there are also talks in the municipal administrative centres, in cooperation with the municipal governments, with the invitation to local authorities of all relevant communities to participate. In addition to talks, an interviewee from Ambio stated that her organization recently conducted workshops to generate awareness among children.

Interviews mentioned that the talks reached only the smallholders living in the communities within the reserve but not the owners or overseers of the private cattle ranches scattered throughout the buffer zone. Only a former Pronatura member said that there had been talks with the members of a rancher association in the early 2010s. In general, the employees of CONANP stated that they would like to do more but needed more resources; thus, in most communities, talks are only given every few years.

Most interviewees from environmental agencies, firefighters, and local authorities agreed that their main aim was to mainstream safe burning techniques. In contrast, there is less interest in alternatives to agricultural burning.

4.2.3. Physical prevention and risk reduction

In addition to the individual precautions of farmers, further measures of physical prevention and risk reduction have emerged since 1998 at the regional and community levels. As the interviews revealed, much of this is carried out by teams of firefighters present in the area: First, there is a professional brigade of the National Forestry Commission (CONAFOR) in the municipal centre of Cintalapa, and second, CONANP temporarily contracts people from the community of Armando Zebadúa as firefighters during the main fire season. Both teams open and maintain several kilometres of fire breaks yearly at strategic points in and around the reserve. However, CONANP operations have diminished in recent years owing to budget cuts.

Personnel from the environmental authorities of the state of Chiapas were also involved until 2021, when they had to hand over the responsibility for fire management to the Secretary of Civil Protection. However, three employees of a small state reserve within the biosphere reserve's buffer zone, Laguna Bélgica, still maintain a fire break along that reserve's boundaries (3 km).

At the community level, fire breaks along the *ejido* boundaries are established and maintained by the local firefighters or the whole community. Usually, there are small daily wages for this activity from projects managed by Ambio or CONAFOR's Payment for Environmental Services program. Until 2019, CONANP also had received finance for these activities from the federal government's Program for Temporal Labor, which was recently discontinued. The resources available through Ambio projects have also diminished. One firefighter lamented:

> *"(…) the firebreak, in reality, it has been neglected, at least in the last years, because there were no resources (…) because it is a hard work of various days."* (free translation, interview 9).

Also, other representatives of governmental organizations, firefighters, and communities emphasized that establishing and maintaining fire breaks in this area is highly laborious because of the karst topography and the risk of subterranean fires, with the need for breaks to be very wide (up to 8 m).

Besides fire breaks, interviewees from the external organizations stated the need to manage the dangerous accumulation of combustible material in the secondary vegetation growing in formerly burnt areas. However, prescribed burning is generally not seen as an option since the natural rainforest vegetation is not fire-adapted. An alternative mechanical removal technique was assessed in 2021 with the cooperation of CONANP, CONAFOR, Ambio, and the state government's environmental secretariat. However, a larger-scale application of this technique would require a considerable number of resources, which are not available at the moment. An employee of one governmental agency complained that prevention is still not made a priority:

> *"It is better to spend on prevention because we have seen that in a fire you can spend up to two million pesos, or even up to six million pesos, for firefighting, but sometimes for prevention, they do not want to give you even half a million."* (free translation, interview 8).

One local firefighter has experimented with the use of invasive plants removed mechanically for handicrafts, such as baskets and furniture, for locals and tourists.

4.2.4. Legal prevention

In Mexico, the NOM-015-SEMARNAT/SAGARPA-2007 (from now on NOM015), decreed in 2007 by the federal environmental and agricultural ministries, defines legal fire use. Only 27.3% of the farmers surveyed and 23.6% of fire users knew this decree. Some interviewees complained that only the environmental authorities were promoting its application, with no efforts by the ministries of agriculture at the federal or state level.

The decree specifies the techniques that need to be applied to burn safely, and it requires farmers to notify when and where they will burn; 69.4% of fire users said that they told the local community authority and 4.2% the local firefighters, making a total of 73.6%; among those knowing the NOM015, the share was 82.4%. 18.1% of all fire users also tell their immediate neighbours. All those who said they knew the NOM-015 stated they established fire breaks before burning.

Also, the reserve's management plan allows responsible fire use within the buffer zone. Contrary to these federal documents, state governors apply a more restrictive approach. In several years, there have even been complete fire bans. According to several intervie-

wees and the focus group discussions, the implementation of these fire bans was weak. However, it motivated some farmers to burn secretly, at night, or without surveillance of the fire, thereby increasing the risk of wildfires. Furthermore, interviewees stated that fire bans damaged the relationship of trust between governmental agencies and local communities and fire users and that the different opinions on fire use have caused discord among the governmental agencies and hampered effective collaboration. This problem has gotten worse in recent years.

In general, implementation of legislation on fire use and other environmental issues is weak in the area; some interviewees had reported cases of arson to federal and state attorneys for environmental protection, but these generally ignored the reports. In recent years, the federal, state, and municipal police have organized joint patrols in parts of the reserve to show their presence and deter people from illegal activities; the effectiveness of these patrols remains unclear.

The following aspects also compromise the rule of law in the area: first, a land conflict between communities in the westernmost part of the reserve, close to the border of Oaxaca; and second, the existence of an unauthorized settlement in one of the core zones, whose inhabitants engage in the illegal planting of marihuana, an issue confirmed by several interviews. One governmental agent who had tried to fight environmental crimes in this area said he had received death threats and chose to work in other areas.

4.2.5. Fire detection and suppression

As with fire prevention, the interviews revealed that fire detection in the area is mainly a shared activity of local communities and governmental agencies. CONANP maintains an observation tower overlooking a large part of the reserve, attended by local firefighters. Furthermore, the reserve registers notifications of thermal hotspots from satellite data and instructs local firefighters or community authorities to check whether there are wildfires or controlled burns. The actors generally considered that this worked well, although the radios and antennas need renewal. In the survey, 85.4% said they knew how to report a wildfire, with 52.2% stating that they would inform local authorities, 28.8% local firefighters, and 3.6% CONANP.

The suppression of fires is perceived by 85.4% to involve the whole community. 21.9% also said that CONANP usually participates in putting out fires, 1.1% mentioned Ambio, and 2.2% the local committees for civil protection, an initiative fostered by the state government. 93.4% of the interviewees were aware of local firefighters in their community. Among them, 62.5% had an outright positive opinion of their work, and a further 14.1% considered that they fulfilled their corresponding tasks. Only 2.9% made negative comments, the main criticism being from Armando Zebadúa, where the CONANP's temporal firefighting team comes from; there, 3.6% of the respondents opposed attendance of the firefighters at fires in other parts of the reserve and demanded that they should attend solely their community. 2.7% of the general respondents were worried about the firefighters' safety, asking for better training to avoid accidents.

The data collected by CONANP for 2017 to 2021 further document the involvement of the brigades of the CONAFOR and the municipal and state secretaries of civil protection. The Mexican Military and the Marines participated in one instance each. External assistance is vital in the case of fires in remote areas such as the core zones and the canyon of the La Venta River. Regarding firefighting in the core zones, firefighters told of the high risk to personal security when attending fires in the area dedicated to growing marihuana.

In the case of fires occurring close to local communities, external experts emphasized the importance of local involvement. The communities sometimes extinguish small fires alone, without external assistance. 28.5% of locals said they had participated in capacity-building for fire suppression, whereas 28.1% said they had not received any training but were interested in obtaining more knowledge. Among men, 48.2% had been trained, and among women 8.8%. However, another 33.6% of women would be interested in capacity-building, compared to 22.6% of men. Of those who had received this training, 52.6% mentioned CONANP, 25.6% Ambio, 7.7% CONAFOR, and 6.4% the local authorities.

External experts and local firefighters underlined the difficulty of fighting fires under the environmental conditions of El Ocote, with the prevalence of subterranean fires:

> *"It is more dangerous because it is silent; it goes on burning, and suddenly a tree falls, or suddenly the flames appear."* (free translation, interview 5).

4.2.6. Restoration of burnt areas

After the widespread fires in 1998, there were efforts to reforest burned areas financed by the federal government and donations from companies; perceptions of the scale of these activities varied widely. No specific follow-up studies that show the long-term impact compared to natural restoration are available, and there are no recent restoration efforts. One employee of the biosphere reserve said in one of the focus group discussions that they had expected regeneration in this tropical rainforest area to take more than fifty years, but they were already seeing great progress in restoration now; another employee of the reserve added:

> *"After 1998, 2003, what was done in terms of restoration was the dispersion of seeds; they stayed there, the rainforest was devastated, and then, twenty years have passed, and when there are observation flights now, it is possible to see that there is already great regeneration. I do not know if because of the seeds that were used, or naturally, but it regenerated, and there are already patches of high tropical evergreen forest again (…)."* (free translation, interview 11).

5. Discussion and conclusions

Due to the ineffectiveness of conventional response-only fire management approaches and the social and environmental problems they cause, researchers and practitioners have sought alternative ways for society to deal with fire. Although many studies have

pointed out problems of prevalent policies and have proposed new approaches and frameworks [16,18], only some studies have documented cases where IFM, as defined by Moore [21], has succeeded over the long term.

The present study has addressed this knowledge gap, analysing the case of the Selva El Ocote Biosphere Reserve, where after widespread and devastating fires of 1998 and 2003, diverse actors have started to work together and seem to have contributed to avoiding any more severe effects of fire on the natural vegetation. Although this study did not provide a counterfactual analysis (given the unique situation of Selva El Ocote in comparison to other areas and the limited data availability for the years before and at the beginning of the reserve's existence), a relationship with management activities is deemed to be highly likely, also considering the upsurge in burnt areas at the same time at the state level, related to meteorological conditions.

According to the interviews, since the outset, fire management in the Selva El Ocote has been chiefly a joint effort of the government and NGOs working together with the communities instead of the antagonism between locals and authorities that has been observed in many other places [13,17]. The approach's success may be partly attributable to the awareness acquired by the local population through the enormous devastation of 1998 when many local inhabitants helped fight the fires and became aware of the value of external support. Success was also favoured by the consciousness, sensitivity, and leadership of a few personalities who determined how fire management was implemented in the reserve, as mentioned by several interviewees.

The main successes of the approach can be summed up the following way, related to the findings outlined in the results section: (1) regarding agricultural fire use, almost all farmers use fire in a controlled manner; to a large part, this is due to cultural fire management, which has been sustained in the area thanks to the overall non-restrictive policy of the most relevant external actors (although there are some exceptions); cultural prevention activities might have contributed as well, although their impact is less clear; (2) according to most of the interviewees, the prevention and risk reduction activities that have been developed after 1998 are the central aspect that has prevented large scale fires from happening again, particularly the considerable efforts to build and maintain firebreaks in strategic locations in the reserve and along community boundaries; (3) wildfires that do occur in the reserve are nowadays detected and suppressed quickly, due to the close cooperation between the corresponding governmental institutions and the local communities, and the preparedness that has been developed at the local level through capacity-building activities.

However, despite the achievements mentioned here, many challenges remain, and the reserve is far from being secure from future devastations by large wildfires. Climate change is increasing fire risk in many parts of the world [48]; this underlines the importance of including a resilience perspective in fire management [18,19]. Invasive plants increase fire risk, particularly in secondary vegetation growing on formerly burnt areas. These areas of secondary vegetation show a significant accumulation of organic matter that may be combustible for large fires in the future. The ideal strategy to tackle this problem would be mechanical fuel management because prescribed fires are inadequate in tropical rainforest areas [49]. Also, the material removed could be used by the local population for firewood [22] or handicrafts.

The scarcity of funding has limited an organized fuel management program. Funding had always been low, and its lack has restricted cultural prevention with almost zero efforts concerning private ranches. Recent reductions have further diminished physical prevention measures. This increases the risk of fires that cost much more to suppress than would have been necessary to spend on prevention, and enlarges the environmental impacts and the security risk for firefighters. This disregard for prevention, which has been reported in other places as well [2], is in utter discord with modern standards of integrated risk management, as stated, for instance, in the Sendai framework of the UN [50].

A further challenge is the intent of some of the actors at the state level to revert to prohibitive and response-oriented approaches in fire management; this not only threatens the good relationship with local communities but has also created frictions among governmental agencies. The proponents of this approach argue that the conditions in Chiapas differ from those in other places, where integrated and cultural fire use have been studied (in contrast to advances in academic literature, precisely from Chiapas [12,22]). Moreover, local experts recounted their experiences that prohibition of controlled burns can be counterproductive and even increase wildfire risk, an issue which has also been reported elsewhere [8]. Given the precarious state of the rule of law in Mexico, a prohibitive approach is unlikely to be feasible. On the contrary, local community agreements have proved to be effective in limiting wildfire risk.

One noteworthy finding of this study was the contradictory impact of the *Sembrando Vida* program. In the beginning, by requesting that registered plots for entering the program were completely free from vegetation, it seems to have caused deforestation (using fire) and some wildfires, as also reported by Pagan et al. [51]. However, subsequently its effect turned to the opposite, prohibiting and causing a widespread decrease in fire use by local farmers. It was beyond the reach of this study to determine what caused these changes, although it might be speculated that a public outcry regarding its initial adverse effects caused the subsequent 180-degree turn in the program's implementation. It remains to be seen whether this will effectively influence fire risk in the reserve, because most farmers already apply safety measures during agricultural burning.

In line with other studies [8,26], the present paper stresses the importance of relating wildfires not only to burning practices of smallholders, but also to fire use on large cattle ranches or by illegal hunters.

A limitation of the present study was not having interviewed any owners of private ranches, due to the difficulty to contact them. Future studies should try to obtain the necessary contacts to talk to ranchers directly. Another obvious limitation was the relatively short time frame of fieldwork, reconstructing previous developments from what interviewees recalled in retrospect. A more longitudinal approach could yield more detailed results on how processes develop. The study also did not consider the areas surrounding the reserve, which may contribute to the dynamics within its boundaries. Moreover, the study is based entirely on social sciences methods, and lacks a more detailed analysis of fire ecology and impacts from wildfires.

Given the new developments in recent years, Selva El Ocote will remain an interesting place for studying fire management, both from a social perspective and regarding fire ecology. Studies are particularly needed relating to the restoration of burnt areas and fuel management feasibility. The results presented here confirm many of the observations on IFM practices made in other places, such as

the importance of prevention and risk reduction activities instead of a suppression-only approach [2,16], and the effectiveness of local fire management efforts and community participation and empowerment [8,12,13,17,22], but this reserve is remarkable in the long-standing experience with IFM approaches and their apparent success, contributing to a "track record" that Wunder et al. [15] speak of.

Although this study was focused mainly on fire management in Selva El Ocote, it highlights the following points that might be researched further, both here and in other areas, particularly in the tropics: (1) the importance of creating a communication base, with regular meetings of stakeholders, particularly between authorities and local communities, (2) the delegation of controlling agricultural fire use to local community assemblies, (3) the importance of preparedness at the local level, (4) the apparent effectiveness of prioritizing preventive measures, and (5) the incipient advances in fuel management adapted to the context of a tropical rainforest ecosystem, including the extraction and use of invasive plants to produce handicrafts. While these issues might be of general interest in many areas, adequate management actions will be different in each specific territory, according to their unique social and environmental characteristics [29].

Another issue this work revealed was that many of the practices carried out by local inhabitants were not based on capacity building and cultural prevention efforts by external actors, but were indeed cultural knowledge held by the communities, as has been outlined elsewhere [11,12,16,29]. Although this study documented the importance of cultural knowledge, it did not obtain more details of specific practices and experiences; this might be a worthwhile undertaking for future studies. However, the study did show the necessity of this kind of knowledge in fire management strategies, adding to calls for implementing intercultural fire management [11,23–25], or, as Tedim et al. put it, of a shared fire governance, that includes empowering local communities to engage in fire management [18].

Integrated fire management in Selva El Ocote might also be enhanced by considering advances from other areas, particularly regarding a consideration of the need to develop adaptive resilience in the face of climate change [18,20,31]; in terms of preparedness, which is a key aspect of IFM [16,21], a gender gap was detected that needs to be addressed as well. Moreover, the challenges outlined throughout the Discussion are a reminder that developing IFM is never a completed task. Rather, it is an ever-evolving process that needs constant revision to ensure its functionality.

Statement regarding the use of AI

During the preparation of this work the authors used the non-generative AI functions of Grammerly (https://app.grammarly.com/) to edit the text's grammar.

CRediT authorship contribution statement

Christoph Neger: Conceptualization, Formal analysis, Funding acquisition, Investigation, Methodology, Project administration, Visualization, Writing – original draft, Writing – review & editing, Data curation. **Laura Patricia Ponce-Calderón:** Data curation, Formal analysis, Investigation, Writing – review & editing. **Lilia de Lourdes Manzo-Delgado:** Data curation, Formal analysis, Investigation, Writing – review & editing. **Mayra Adriana López-Madrid:** Data curation, Formal analysis, Investigation, Visualization, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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